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Relating Redox Properties of Polyvinylamine-g-TEMPO/laccase Hydrogel Complexes to Cellulose Oxidation

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KEYWORDS

Cellulose oxidation, TEMPO, cyclic voltammetry, laccase, redox polymer,

EQCM-D, redox hydrogel

ABSTRACT

The structure and electrochemical properties of adsorbed complexes based on mixtures of polyvinylamine-g-TEMPO (PVAm-T) and laccase were related to the ability of the adsorbed complexes to oxidize cellulose. PVAm-T10 with 10% of the amines bearing TEMPO moieties (i.e. DS= 10%), adsorbed onto gold-sulfonate EQCM-D sensor surfaces giving a hydrogel film, 7 nm thick, 89% water and encasing laccase (200 mM) and TEMPO moieties (33 mM). For DS values >10% all of the TEMPOs in the hydrogel film were redox-active, in that they could be oxidized by the electrode. With hydrogel layers made with lower DS PVAm-Ts, only about half of the TEMPOs were redox-active; 10% DS appears to be a percolation threshold for complete TEMPO-to-TEMPO electron transport. In parallel experiments with hydrogel complexes adsorbed onto regenerated cellulose films, the aldehyde concentrations increased monotonically with the density of redox-active TEMPO moieties in adsorbed hydrogel. The maximum density of aldehydes was 0.24 $\mu mol/m^2$, about ten times less than the theoretical concentration of primary hydroxyl groups exposed on crystalline cellulose surfaces. Previous work showed that PVAm-T/laccase complexes are effective adhesives between wet cellulose surfaces when the DS>10%. This work supports the explanation that TEMPO-to-TEMPO electron transport is required for the generation of aldehydes necessary for wet adhesion to PVAm.

INTRODUCTION

We recently described a protein/synthetic polymer complex that spontaneously adsorbs onto cellulosic surfaces, selectively oxidizes the cellulose, and then forms covalent grafts to the cellulose surface. [1] The grafted protein/polymer complex formed very strong adhesive joints between wet cellulose surfaces. Other than oxygen, all of the reactive components were built into the protein/polymer complex, giving reagentless polymer grafting to cellulose at neutral pH and room temperature. The auto-grafting complexes are formed by mixing the redox protein laccase with polyvinylamine-g-TEMPO (PVAm-T). Figure 1 shows the structure of PVAm-T and illustrates the oxidation and subsequent grafting chemistry. We proposed that the TEMPO moieties, grafted to PVAm, served to transport electrons from the cellulose surface to the laccase active sites via a series of TEMPO-to-TEMPO electron jumps - see Figure 2A. Although reasonable, our proposed mechanism was based mainly upon adhesion measurements involving a number of control experiments.

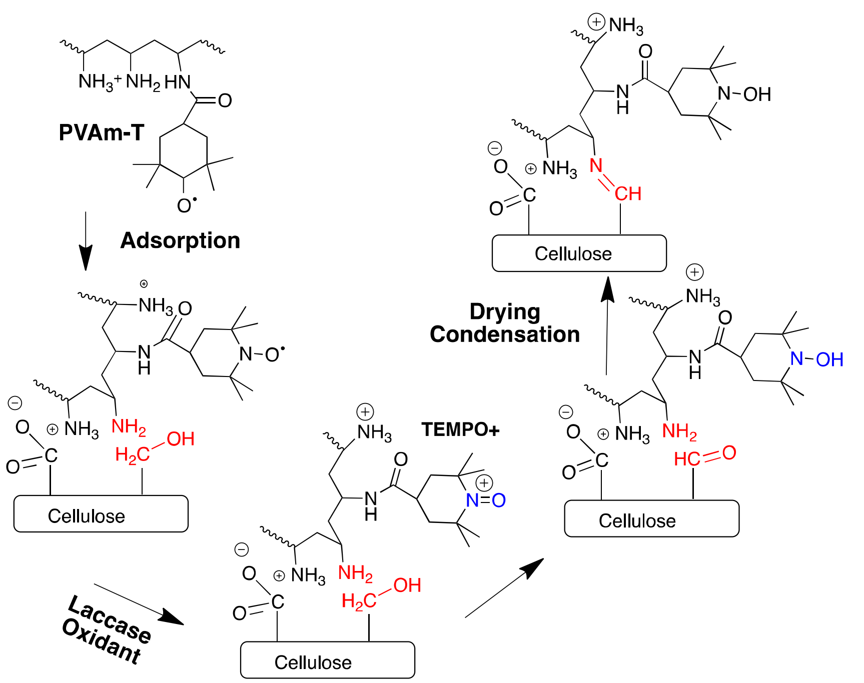
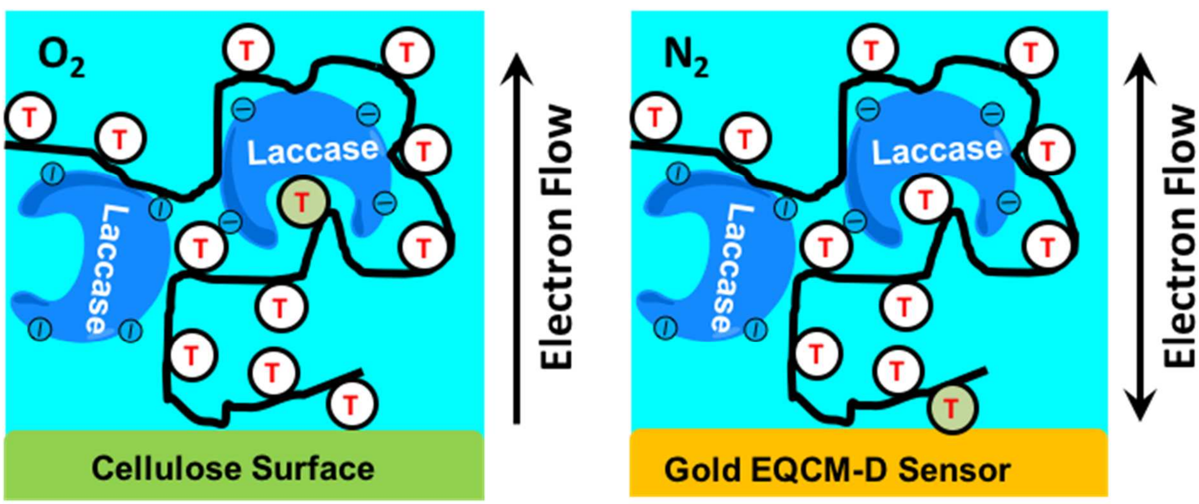


Figure 1 PVAm-T adsorbs onto cellulose, oxidizes cellulose and, with drying forms covalent imine and aminal linkages to cellulose. [2]

This paper describes the second half of a project with the overall objectives of: A) characterizing PVAm-T/laccase complex structure in solution and on cellulose; B) determining the influence of TEMPO content (i.e. the DS) on the fraction of TEMPO groups participating in electron transport; and, C) establishing the relationship between TEMPO-to-TEMPO electron transport in the PVAm-T/laccase complex and the surface density of aldehydes formed on cellulose. In the recently published first half, we showed that cyclic voltammetry could be used to measure the fraction of TEMPO moieties participating in electron transport (i.e. were "redox-active") when the PVAm-T was adsorbed on gold electrodes.[3] Only about a third of TEMPOs were redox-active when PVAm-T was deposited as adsorbed multilayers with polystyrene sulfonate. Furthermore, we showed that the redox properties of our PVAm-T polymers were consistent with the literature describing other TEMPO modified water-soluble polymers.

Herein we relate measurements from two physical systems shown schematically in Figure 2. With PVAm-T/laccase adsorbed onto cellulose substrates, we measured the coverage (μmol/m^2) of cellulosic aldehyde groups generated by oxidation. With complexes adsorbed onto the gold-sulfonate electrode surfaces, Figure 2, we measured the coverage of redox-active TEMPO moieties in the adsorbed complexes. Thus, we were able to relate the redox-active TEMPO contents in the adsorbed PVAm-T/laccase hydrogel layer to the extent of cellulose oxidation. In addition, this work confirms the importance of TEMPO-to-TEMPO electron transport when oxidizing cellulose with adsorbed PVAm-T/laccase complexes.



A: Cellulose Oxidation **B: Redox-active TEMPO Determination**

Figure 2 Schematic illustration of the two physical systems used herein. Cyclic voltammograms, measured on gold, were used to estimate the density of redox-active TEMPOs in the cellulose oxidation experiments.

EXPERIMENTAL

Materials. Polyvinylamine (PVAm) (Mw 45 kDa, hydrolysis degree 75%), obtained from BASF Canada (Lupamin 5095), was dialyzed against deionized water and freeze-dried for storage. Sodium 3-mercapto-1-propanesulfonate (MPS), 4-carboxy-TEMPO, 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide (EDC), N-hydroxysulfosuccinimide sodium salt (sulfo-NHS), Laccase from Trametes versicolor (catalog no. 51639), were all purchased from Sigma-Aldrich. Bodipy FL hydrazide (catalog no. D2371), was purchased from Life Technologies. Other salts for buffer preparation were purchased from Caledon Laboratories. All solutions were made with deionized water (18.2 MΩ cm^{-1}, Barnstead Nanopure Diamond system). Cellulose membranes were cut from regenerated cellulose dialysis tubing with 12-14 kDa molecular weight cut-off (MWCO) was purchased from Spectrum Laboratories. The dry membranes were 70 micrometers thick with a roughness of about 60 nm. The tubing was boiled for 30 min and washed with deionized water to remove preservatives before use.

PVAm-T Synthesis and Characterization. PVAm-T was synthesized through EDC coupling reaction and the PVAm-T compositions are given Table S1. In a typical experiment, 55 mg 4-carboxy-TEMPO was dissolved in 100 mL deionized water and 157 mg EDC and 170 mg Sulfo-NHS were added. The solution was stirred for 20 min at pH 5.5 at room temperature. PVAm solution (100 mg PVAm in 40 mL deionized water) was gradually added into the above solution, and the solution pH was maintained at 7 for 4 h with 0.1 M HCl and 0.1 M NaOH. The product was dialyzed (MWCO 12-14 kDa) against deionized water for two weeks. The final products were freeze-dried and stored in a desiccator. The amine contents of PVAm-T were determined by potentiometric and conductometric titrations. The grafted TEMPO contents were calculated from the change in amine contents.

PVAm-T/laccase Complex Preparation. PVAm-T and laccase stock solutions (1g/L) were prepared with 50 mM acetate buffer at pH 5 and the solutions were passed through 0.45 µm filters before use. In a typical experiment, 2.5 mL 1 g/L PVAm-T13 solution was first diluted into 30 mL acetate buffer, and the diluted solution was stirred 30 min before taking any further actions. 5 mL 1 g/L laccase solution was then slowly dropped (~2 min/mL) into the above dilute PVAm-T13 solution with magnetic stirring. For cyclic voltammetry studies nitrogen was continuously bubbled through the solution to prevent laccase catalyzed oxidation. In the cellulose oxidation experiments, oxygen was continuously bubbled into the PVAm-T/laccase complex dispersion.

Phase Behavior Measurements. The phase behaviors of the PVAm-T/laccase complexes were obtained by measuring the optical density at 500 nm with a Backman DU800 UV-vis spectrophotometer. In a typical experiment, 3 mL PVAm-T17/laccase complex solution was aged 1h, and then added to a 10 mm silica cuvette, the optical density at 500 nm was measured immediately.

Dynamic Light Scattering (DLS). The size of PVAm-T/laccase complexes was measured by DLS (Brookhaven dynamic light scattering instrument) at a 90^{o} scattering angle. A Melles Griot He-Ne laser with a wavelength of 633 nm served as the laser source, and the data were analyzed using the CONTIN method.

Electrophoresis measurements. The electrophoretic mobility of PVAm-T/laccase complexes were measured by a ZetaPlus Analyzer (Brookhaven Instruments Corp.) with PALS (phase analysis light scattering) Software Version 2.5.

Gold-sulfonate EQCM-D Sensors. Q-Sense E4 QCM QSX 301 gold sensors with a surface roughness specification of 3 nm (RMS) were cleaned by piranha solution at 75 °C for 5 minutes and then thoroughly rinsed with deionized water. Note piranha is dangerous and appropriate precautions must be taken when working with it. The cleaned Au chip was immersed into 3 mM sodium 3-mercapto-1-propanesulfonate ethanol solution for 24 hours, followed by rinsing with ethanol and then by deionized water.

Adsorption of PVAm-T/laccase Complexes on Gold-sulfonate Sensor Surfaces. The adsorption of PVAm-T/laccase complexes was monitored by quartz crystal microbalance with dissipation (QCM-D). All solutions were made with 50 mM acetate buffer at pH 5. In a typical experiment, 1g/L laccase and 76.9 mg/L PVAm-T13 solutions were purged with nitrogen for 30 min after which 5 mL laccase solution was slowly dropped (~2 min/mL) into 32.5 mL PVAm-T13 solution under magnetic stirring and nitrogen bubbling during the mixing process. After aging the mixture for a given time under continued nitrogen bubbling at room temperature, the PVAm-T/laccase dispersion was pumped into the QCM at a flow rate of 0.100 mL/min for a fixed time (usually one hour) followed by rinsing with buffer. All measurements were performed at 23 ^{o}C.

Ellipsometry. The mass coverage of dried complex films on EQCM-D gold-sulfonate surfaces was estimated from ellipsometry. The thickness of the air-dried PVAm-T/laccase complex films was measured using a Nanofilm EP3W imaging ellipsometer (Accurion Inc., Germany) fitted with a wavelength 658 nm solid state laser. The ellipsometry data was analyzed using the modeling software EP4 (Accurion Inc.).

The refractive index of the PVAm-T/laccase complex films was calculated as the average refractive index of PVAm-T and laccase. The measurement of PVAm-T refractive index was reported previously. [3] The laccase refractive index was measured from thick spin coated films prepared from 0.1 wt % aqueous laccase solution at 500 rpm on piranha-cleaned silicon wafers. The ~100 nm thick laccase films were annealed at 50 ^{o}C overnight after which the refractive index was determined by ellipsometry. The refractive index of laccase was 1.422 ±0.044, where the error estimate was based on three measurements. Finally, the coverage (dry mass/ area) of PVAm-T/laccase complex films were estimated from the incremental ellipsometric thickness and the average density of PVAm-T (1.08 g/mL, assumed to be the same as PVAm) and laccase (1.20 g/mL).

Electrochemical Quartz Crystal Microbalance with Dissipation (EQCM-D). The EQCM-D measurements were carried out with an electrochemical QCM-D cell (E401 electrochemical model from Q-Sense, Sweden). A QSX301 gold chip served as the working electrode, a platinum thin slice in the E401 electrochemical model served as the counter electrode and a low leakage "Dri-Ref 2SH" Ag|AgCl electrode (World Precision Instruments, USA) served as the reference electrode. Electrochemical measurements were performed with a PalmSens potentiostat (PalmSens, Amsterdam, Netherlands) in a conventional three-electrode system. After absorbing PVAm-T/laccase complexes, the pump was stopped before running electrochemical experiments. 50 mM sodium acetate buffer with pH 5 served as the solvent for both polymer and laccase solutions as well as the rinsing buffer. All measurements were performed at 23^{o}C.

Cellulose oxidation. Circular disks with a diameter of 10 mm were punched from cleaned regenerated cellulose membranes. In a typical oxidation experiment, PVAm-T17 and laccase were premixed. The final concentration of PVAm-T17 was 67 mg/L and laccase 133 mg/L. 4 cellulose disks were immersed in the above solution and the solution was stirred with oxygen purging for 1h. Both the PVAm-T17 and laccase solutions were filtered through a syringe filter (0.45 μm) before mixing. The resulting oxidized cellulose disks were washed thoroughly with deionized water.

Aldehyde Quantitation. The aldehyde concentration on the surface of the cellulose disks was estimated from fluorescence measurements of the hydrazide dye Bodipy FL. The hydrazide groups conjugate with aldehydes in water giving the structure shown in Figure 9. The oxidized cellulose disks were immersed in 10 μM solution of the fluorescent Bodipy FL hydrazide in methanol/water (methanol: H_2O = 5:1) at room temperature for 8 h. The disks were washed by immersion in methanol for over 5 h with stirring, and then in acetone for another 5 h. The washing solvents were refreshed once per hour.[4] The washed disks were dried by nitrogen at room temperature. The fluorescence intensities of the dried disks were measured with a ChemiDoc MP system (Model Universal Hood Ⅲ, Bio-Rad Laboratories, Inc.) using the "rhodamine mode". The excitation wavelength was 520 ± 50 nm and the fluorescent emission filter was 605 ± 50 nm.

Calibration curves were made using unmodified cellulose disks onto which 5 μL drops of Bodipy FL hydrazide methanol solution (0-10 μM) were placed and allowed to dry. Fluorescent intensity was measured using the ChemiDoc and the resulting calibration curve is shown in Figure S1.

RESULTS

Pvam-T/Laccase Complex Phase Behavior at Long Times. A series of PVAm-T copolymers was prepared by conjugating 4-carboxy-TEMPO to polyvinylamine-co-N-vinylformamide copolymers. The PVAm-T detailed structure is shown in Figure S2 and the detailed copolymer compositions are summarized in Table S1 of the supporting information file. The specific copolymers are designated PVAm-TDS, where DS is the percentage of amine groups conjugated to TEMPO.

Polyelectrolyte complexes were prepared by mixing PVAm-T solutions with solutions of the enzyme laccase. The experiments were performed in 50 mM acetate buffer at pH 5. Under these conditions PVAm-T is positively charged and laccase is negatively charged [5] (-1.3 meq/g). Figure 3 shows the phase boundaries of the PVAm-T13/laccase complexes. PVAm-T13 was chosen for this detailed analysis because it is one of the most promising wet cellulose adhesives when used with laccase.[1] The phase boundaries of the PVAm-T13/laccase complex were assigned based on the following criteria: "Soluble Complex" refers to solutions with absorbance values less than 0.02 at 500 nm; "Stable Colloid" refers to dispersions with absorbance values >0.02 but with no visible precipitates; and, colloidal phases that have visible precipitates after 24 h are designated as "Unstable Colloid". The phase boundaries in Figure 3 were based on a large data set – see Figure S3 in the supporting information.

The black diamonds and squares in Figure 3 denote specific compositions and the numbers beside the symbols give the corresponding electrophoretic mobility ($m^2/Vs \times 10^{-8}$)/diameter (nm). The diamonds denote conditions with excess laccase where the colloidal complexes were negatively charged. By contrast, the compositions denoted by squares had excess PVAm-T and were positively charged.

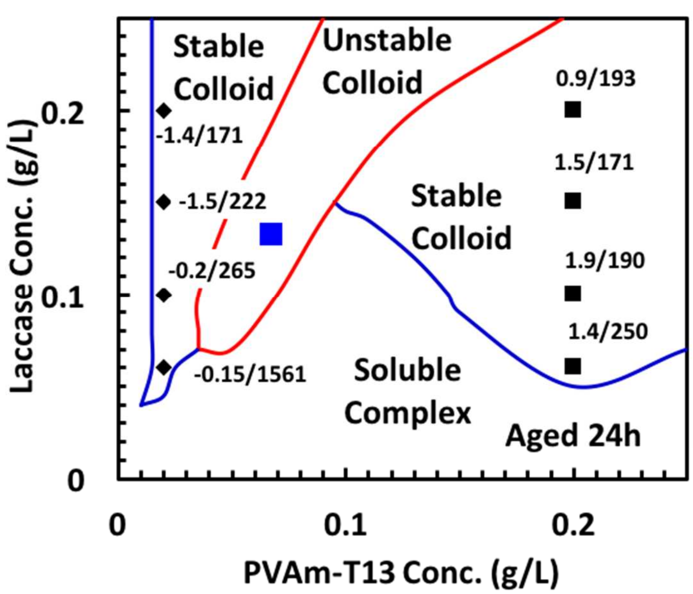


Figure 3 Phase boundaries for mixtures of PVAm-T13 and laccase in 50 mM acetate buffer at pH 5. Laccase solutions were slowly dropped into PVAm-T13 solution and the mixtures were aged 24 h before characterization. The numbers beside the data points are electrophoretic mobility (10^{-8} m^2/Vs) and mean diameter (nm) values. The blue

square indicates the composition used for many of the experiments described below. The data used to define the phase boundaries are shown in Figure S3. The net negative charge content of laccase, expressed as an equivalent weight, is 4.5 times greater than the equivalent weight of PVAm-T13 at pH 5.

Finally, many of our experiments, described below, were conducted using mixtures of 67 mg/L PVAm-T + 133 mg/L laccase; this composition is plotted as a large square in Figure 3 centered in the "Unstable Colloid" domain. We focused on this composition because it was found to give the maximum wet cellulose adhesion in our initial work. [1] The charge content of PVAm-T13, expressed as an equivalent weight, was 171 Da. Under these conditions laccase, with an isoelectric point of 3,[6] was negatively charged with an equivalent weight of 769 Da. [5] Therefore the mixture of 67 mg/L PVAm-T13 + 133 mg/L laccase has approximately 2.3 more cationic ammonium groups compared to the excess anionic groups on laccase. The time-dependent complex properties are presented below. The corresponding electrophoretic mobility of the complex, shortly after preparation, was +1.1 x10^{-8} m^2/Vs

In summary, the solution behavior of PVAm-T/laccase mixtures display the classic features of polyelectrolyte complexes formed between oppositely charged water-soluble polymers. For example, the phase domains in Figure 3 are similar to complexes formed between PVAm and carboxymethyl cellulose. [7]

Pvam-T/Laccase Complex Time-dependent Solution Properties. In our preliminary study, we reported that the enzyme activity of PVAm-T14/Laccase (Figure 7 in reference [1]) decreased by 80% in 24 hours. The enzyme activity assay involved a low molecular weight substrate that did not require a TEMPO mediator. We proposed that PVAm-T chains slowly block active sites as the complex structure evolved. Figure 4 shows the evolving phase boundaries for a series of PVAm-TDS (67mg/L)/laccase (133 mg/L) complexes. The initial electrophoretic mobility values for the complexes are shown in Figure S4 and the complete data set for the phase boundaries is shown in Figure S5. The results in Figure 4 showed that most of the compositions were present as slowly aggregating colloids – the higher the TEMPO DS, the lower the useful working time. In the extreme, complexes with TEMPO DS>18% precipitated quickly, limiting their ulitity as reactive adhesive primers. By contrast, PVAm-T13/laccase took about 14 hours to precipitate. Figure S6 shows that during the aging process, the electrophoretic mobility values decreased, whereas the apparent average particle size increased.

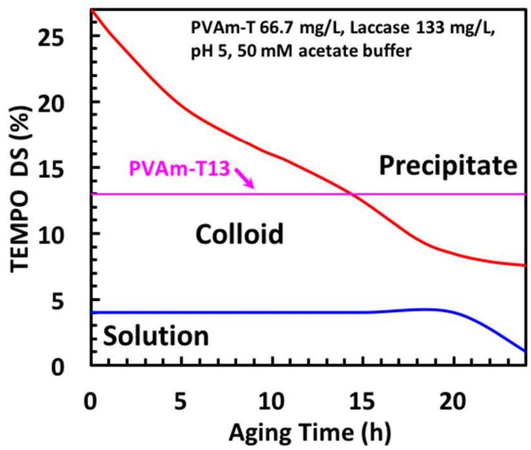


Figure 4 The influence of TEMPO DS and aging time on the PVAm-T phase. The data points used to construct the boundaries are shown in Figure S5.

We evaluated PVAm-T17/laccase aging by monitoring the properties of adsorbed layers on a gold-sulfonate QCM-D sensor surface. Figure 5A shows the QCM-D adsorption properties as a function of the pre-deposition aging time of the PVAm-T17/laccase complex dispersion. There was a large decrease in both the magnitude of the frequency shift and the dissipation over the first two hours of aging. The corresponding dry adsorbed masses, measured by ellipsometry, also decreased with aging time; the results are plotted as functions of the corresponding wet masses in Figure 5B. This is unusal behavior and these aging properties influence the redox properties.

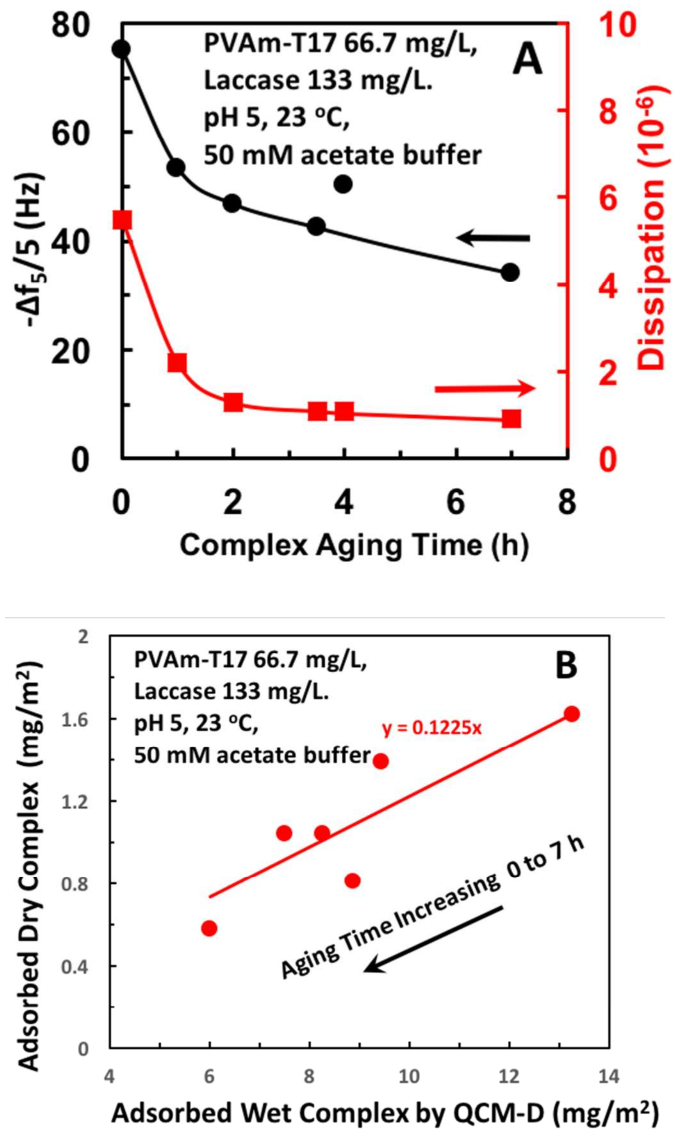


Figure 5 A: Effect of pre-deposition, complex aging time on the properties of complex measured adsorbed on gold and measured by EQCM-D. B: Comparing dry complex coverage, measured by ellipsometry to wet coverage measured by QCM-D.

Cyclic voltammetry was used to measure the quantity of redox-active TEMPO in PVAm-T/laccase complexes adsorbed on the gold-sulfonate EQCM-D sensor surfaces. Figure S7 illustrates the method used to extract the oxidation charge from cyclic voltametry curves. The oxidation charge for a single sweep is shown as a function of the pre-deposition complex aging time in Figure 6; see Figure S8 for the corresponding cyclic voltammetry curves. The redox-activity of the PVAm-T/laccase adsorbed to the gold electrode decreased dramaticaly with the time that the complexes were aged before deposition. Taken together, the results in Figure 5 and Figure 6 suggests that with aging, the quantity of adsorbed PVAm-T/laccase complex decreases, giving a lower redox response.

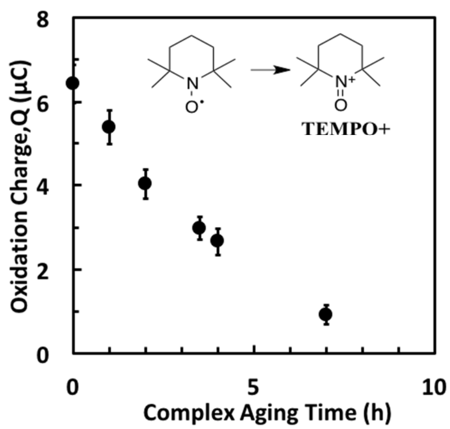


Figure 6 The total electron flow (i.e. the oxidization charge) resulting from TEMPO oxidation as a function of pre-deposition complex aging time for the electrode surfaces corresponding to Figure 5.

The Influence of TEMPO DS in Pvam-T/Laccase Complexes on Redox Activity and Cellulose Oxidation. The goal of these experiments was to measure the fraction of redox-active TEMPO moities in PVAm-T/laccase complexes as a function of the TEMPO DS values for PVAm-T. This information is used relate the electrochemical properties of adsorbed PVAm-T/laccase complexes to cellulose oxidation. The key requirement for redox-activity is electron transport between neighboring TEMPOs in the PVAm-T/laccase hydrogel films. Redox-active TEMPOs must be within ~0.6 nm of a neighboring TEMPO for electron hopping to occur. [8] Two measurement types were required to determine redox activity, the amount of PVAm-T/laccase complexes adsorbed onto the gold-sulfonate EQCM-D sensors, and the total electron flow between the adsorbed complexes and the gold-sensor surface during the oxidation half of cyclic voltametry sweeps. The coverage (mass/area) of adsorbed complexes was measured by EQCM-D frequency changes for wet mass and by ellipsometry for dry mass – the coverage results are now presented.

Figure 7 shows the influence of TEMPO DS on the frequency and dissipation responses to complex adsorption. For soluble PVAm-T/laccase complexes with TEMPO DS <6%, the adsorption reached a steady-state within 30 minutes. Furthermore, the dissipation was low ($\Delta D < 10^{-7} \Delta f_5/5$) and the set of overtone curves showed good agreement (see Figure S9), suggesting that PVAm-T (DS<6%) /laccase complexes were not highly hydrated. By contrast, PVAm-T/laccase complexes with 20%>TEMPO DS>6%, showed a slow continual frequency drop after one hour exposure to flowing complex dispersion (see Figure S10), indicating significant hydration. Finally, complexes made with PVAm-T20 and PVAm-T25 gave large frequency shifts and dissipation values, suggesting deposition of aggregated colloidal complexes.

The corresponding dry coverages were measured by ellipsometry and the results are summarized in Table 1 together with EQCM-D results. Figure S11 shows a linear relationship between the dry film coverage, Γ_D, and the wet film coverage, Γ_W, estimated from the Sauerbrey equation; the wet mass was about 5 times greater than the dry mass. The corresponding redox measurements are now presented.

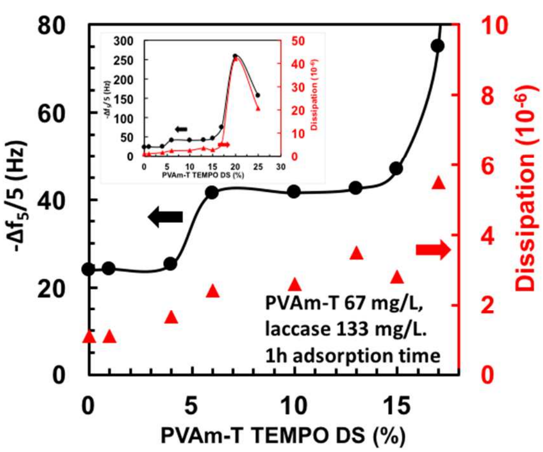


Figure 7 The influence of TEMPO DS on the adsorption of PVAm-T/laccase complexes on gold-sulfonate EQCM-D sensors. PVAm-T (67 mg/L) was mixed with laccase (133 mg/L) under a nitrogen blanket. The inset shows an expanded range of DS values. The reported values were obtained after the mixtures were pumped through the EQCM-D sensor for 1 h followed by with a buffer rinse.

Table 1 Properties of PVAm-T_{DS} (67 mg/L)/laccase (133 mg/L) adsorbed on a sulfonated gold EQCM-D sensor in 50 mM acetate buffer at pH 5. *

	$\Delta f_5/5$ (Hz)	D (10^{-6})	Γ_W (mg/m^2)	Γ_D (mg/m^2)	Q (μC)	Γ_{TRA} (μmol/m^2)	f_{RA}
PVAm	-24.0	1.1	4.2	0.232	0	0	0
PVAm-T1	-24.2	1.1	4.3	0.464	0.08 ± 0.08	0.011	0.5
PVAm-T4	-25.2	1.7	4.5	0.464	0.22 ± 0.16	0.029	0.5
PVAm-T6	-41.5	2.4	7.3	0.58	0.55 ± 0.20	0.073	0.5
PVAm-T10	-41.6	2.6	7.4	0.812	1.20 ± 0.15	0.158	0.6
PVAm-T13	-42.5	3.5	7.5	0.928	3.13 ± 0.18	0.413	1.1
PVAm-T15	-46.9	2.8	8.3	1.044	4.03 ± 0.30	0.532	1.1
PVAm-T17	-75.0	5.5	13	1.624	6.42 ± 0.45	0.848	1.0
PVAm-T20	-259	42	46	7.888	30.2 ± 1.93	-	-
PVAm-T25	-158	20.7	28	4.872	20.3 ±1.13	-	-

*Δf_5 is the frequency change of the 5th overtone; D is the corresponding dissipation values; Γ_W is the wet mass coverage calculated with the Sauerbrey equation; Γ_D is the dry mass coverage adsorbed complex estimated from ellipsometric measurements; Q is the oxidation charge; Γ_{TRA} is the coverage of redox active TEMPO calculated from Q and Γ_D; and, f_{RA} is the fraction of redox-active TEMPO moieties. QCM measurements were made after the complex dispersions were pumped through the cell for 1h followed by a short buffer rinse. The electrochemical measurements were made immediately following the QCM data collection.

Cyclic voltammetry experiments were performed immediately after the QCM-D measurements. An example voltammogram is shown in Figure S7. The total electron flow in the oxidative sweeps, the Q values, are given in Table 1. The corresponding coverage of redox-active TEMPO, Γ_{TRA} (μmol/m^2), was calculated from the dry mass coverages, Γ_D, of complex on the electrode surface with the corresponding Q value from the cyclic voltammetry. The sample calculation in the Supporting Information highlights the many assumptions. Note that the Q values, and therefore Γ_{TRA} and f_{RA} values, were approaching the sensitivity of our instrument for the three lowest TEMPO contents.

The final column in Table 1 is f_{RA}, the fraction of TEMPO groups that are redox-active. The results fall into two groups. Up to a TEMPO DS of 10, the f_{RA} values are around 0.5, whereas f_{RA} = 1 for high DS values. We propose the transition from f_{RA} = 0.5 to f_{RA} 1 between DS 10 and 13% as percolation threshold; for DS < 10% only TEMPOs in direct contact with the electrode are redox active whereas for DS >10%, all the TEMPO moities can particpate in electron transport.

The Γ_{TRA} values, tabulated in Table 1, are plotted as a function of PVAm-T DS values in Figure 8. Shown in the same figure are published wet-adhesion forces required to delaminate cellulose membranes with a grafted layer PVAm-T also as a function of DS. [1] Consider first the coverage of redox-active TEMPO, the black triangles in Figure 8. Γ_{TRA} increases non-linearly with TEMPO DS. Clearly there is a significant change at DS 10, reminiscent of a percolation threshold. The older adhesion data shown in Figure 8 also showed a non-linear dependence on PVAm-T TEMPO content. There was little adhesion advantage in using TEMPO DS values greater than 10 in spite of the dramatic increases in the density of redox-active TEMPO.

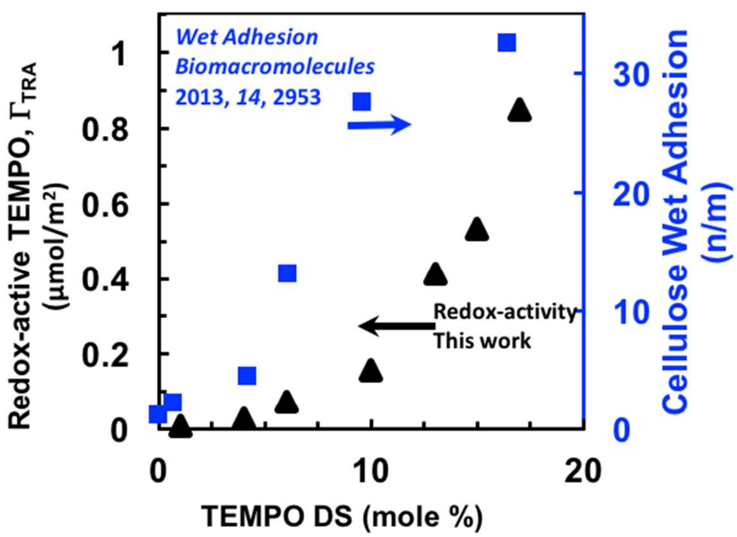


Figure 8 The coverage of redox-active TEMPO in PVAm-T/laccase complexes on gold sensors, determined by cyclic voltammetry. The fraction of redox-active groups was calculated as the ratio of the redox-active TEMPO coverage to the total TEMPO coverage, estimated from the ellipsometric characterization of the dried complex films. PVAm-T (67 mg/L) and laccase (133 mg/L) buffer solutions were mixed, and immediately added into the EQCM-D at a flow rate of 0.1 mL/min at 23 ^{o}C. The complex dispersions were pumped across the sensor surface for one hour followed by rinsing with acetate buffer. All solutions were stored under a nitrogen blanket.

Linking PVAm-T Oxidation and Grafting to Cellulose to the Redox Behavior on Gold-Sulfonate. Our PVAm-T polymers were designed to adsorb onto cellulose and promote the oxidation of primary alcohols to aldehyde groups. With drying, the PVAm-T amine groups form covalent imine bonds with aldehydes, grafting PVAm-T to cellulose surfaces (see structures in Figure 1). [9] The goal of the following experiment was to determine the relationship between the density of aldehyde groups on cellulose to the density of redox-active TEMPO moieties in the complexes that oxidized cellulose. We measured the density of cellulosic aldehyde groups by reacting the oxidized cellulose, before drying, with Bodipy FL hydrazide dye – see inset in Figure 9. The reactive dye forms covalent hydrazone bonds with the aldehyde groups on cellulose. Regenerated cellulose films were oxidized with our series of PVAm-TDS/laccase complexes and the density of aldehyde groups generated are shown as a function of the TEMPO DS in Figure 9. There is a linear relationship between aldehyde density and PVAm-T DS, which is surprising in view of the nonlinearity of the adhesion and redox-active data in Figure 8.

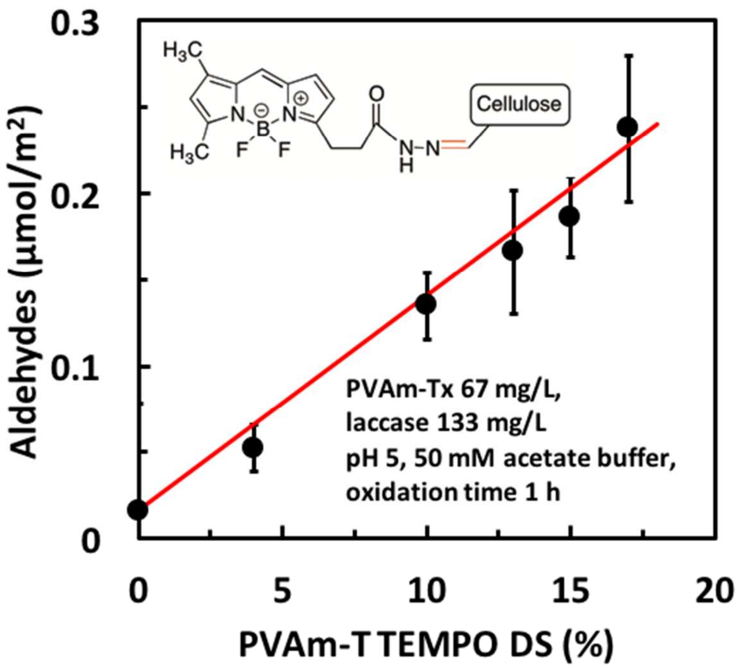


Figure 9 The influence of PVAm-T DS on the density of aldehyde groups on cellulose produced by adsorbed PVAm-T/laccase complexes. Oxidation experiments were performed over 1 hour in 50 mM acetate buffer solutions containing 67 mg/L PVAm-T, 133mg/L laccase, in pH 5 buffer. The error bars represent the range of triplicate samples.

DISCUSSION

PVAm-T is a unique polymer in that it carries built-in TEMPO moieties that promote cellulose oxidation, [2] enabling the primary amines on the PVAm backbone to react with cellulosic aldehydes forming covalent imine and aminal bonds [9] upon drying – see Figure 1. Herein we have probed the details of the PVAm-T/laccase complex structure and redox activity as a function of degree of TEMPO substitution of PVAm-T. By grafting TEMPO onto PVAm and forming a cationic hydrogel complex with laccase, the redox active agents are concentrated at the cellulose/water interface. This is illustrated by the properties of PVAm-T10/laccase complexes, tabulated in Table 2. The PVAm-T10/Laccase complexes deposits on the gold-sulfonate surface to give a hydrogel coating that is 7.4 nm thick, containing 89% water.

The TEMPO moiety content of the adsorbed hydrogel layer is expressed three ways in Table 2: 1) the molar coverage, which is the total TEMPO content divided by the electrode area; 2) the TEMPO concentration, based on the assumption that the TEMPOs are uniformly distributed in the hydrogel layer; and, 3) the corresponding volume per TEMPO moiety. The TEMPO moiety concentration of 33mM is larger than the reported molecular TEMPO concentrations used to oxidize cellulose with laccase in solution, 3-26 mM [10] or 0.13 mM,[1] suggesting that less mobile, grafted TEMPO is less effective than free TEMPO as a oxidation mediator.

Table 2 Properties of PVAm-T10/laccase complex hydrogel on the EQCM-D gold sulfonate surface.

Properties	**PVAm-T10/laccase Adsorbed Complex***
Adsorbed layer thickness	7.4 nm
Water content	89% wt/wt
Molar coverage of TEMPO moieties	0.24 μmol/m^2
TEMPO moiety concentration in hydrogel layer	33 mM
Volume per TEMPO	50 nm^3
Laccase concentration in hydrogel layer	200 mM

* Calculated using Γ_W and Γ_D values from Table 1 and equivalent weights from Table S1. Assumptions include: Sauerbrey model gives accurate wet mass; mass ratio of PVAm-T/laccase in complex is equal to feed ratio; and, density of hydrogel is 1g/mL.

The laccase molar concentration in the adsorbed gel layer was high (200 mmol/L) and nearly 10 times higher than the TEMPO concentration. We showed previously that there is a steady decrease in oxidation efficiency, as judged by wet adhesion, when the enzyme concentration in the complexes is decreased (see Figure 5 in [1]). The requirement of high enzyme and TEMPO concentrations in the adsorbed complex is not surprising because both the laccase molecules and the TEMPO moieties have limited mobility when trapped in the hydrogel. We have proposed that TEMPO-to-TEMPO electron transport is required for cellulose oxidation with PVAm-T/laccase hydrogel. However, redox-active TEMPOs must be within ~0.6 nm of a neighboring TEMPO moiety for electron hopping to occur. [8] If we assume a uniform distribution of TEMPOs in the hydrogel layer, the average volume per TEMPO in the complex is 50 nm^3, corresponding to an average TEMPO-to-TEMPO nearest neighbor distance of approximately $50^{1/3}$ = 3.7 nm, six times greater than 0.6 nm. However, about 90% of the adsorbed complex is water (Table 2). Therefore, instead of being uniformly distributed, the TEMPOs are concentrated along the PVAm chains, facilitating TEMPO-to-TEMPO electron transport along PVAm-T chains and at chain-chain junctin points. For example, the TEMPO-to-TEMPO spacing on fully extended PVAm-T10 with a uniform TEMPO distribution is about 2.5 nm (20 x 0.126nm). Therefore along-the-chain electron tranport seems more likely than inter-segment transport. We now link the electrochemical properties, measured for PVAm-T/complexes adsorbed on gold, to cellulose oxidation.

Figure 2 shows schematically the two physical systems we are relating - PVAm-T/laccase complex adsorbed on cellulose and PVAm-T/laccase complex adsorbed on gold-sulfonate. Our analysis is based on two severe assumptions. First, we assume that the structure and the mass density of the adsorbed complex is the same on both gold-sulfonate and cellulose. Second, we assume that measured content of redox-active TEMPOs on gold is equal to the redox-active TEMPO content on cellulose. This is equivalent to assuming the electron flow pathways from the TEMPOs to the gold are equivalent to the pathways from the TEMPOs to the laccase active sites.

Based on these assumptions, we correlated the redox-activity determinations in Figure 8 to the cellulose oxidation results in Figure 9. Figure 10 shows that the measured aldehyde densities on the cellulose membrane surfaces increased monotonically with our estimates of the redox-active TEMPO coverage. Furthermore, the densities of aldehydes on cellulose are of the same order of magnitude as the redox-active TEMPO densities. However, the yield of aldehyde groups was below the theoretical maximum. Based on Gray's AFM analysis of cellulose crystal structures, the maximum surface density of aldehyde groups on a crystalline cellulose surface is 2.7 $\mu mol/m^2$, 10 times higher than our measured densities in Figure 10.[11] In order to oxidize cellulose, a TEMPO moiety must be in direct contact with a cellulose primary hydroxyl [12] and there must exist a redox pathway between the surface TEMPO and the active site on a laccase. Considering these restrictions, it is remarkable that we achieved ~ 10% conversion to aldehydes.

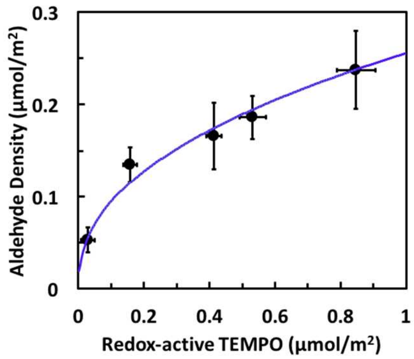


Figure 10 The relationship between the redox-active TEMPO density in PVAm-T/laccase, measured on gold, and the aldehyde density produced by PVAm-T/laccase adsorbed on cellulose.

Finally, coating wood pulp fibers with a PVAm-T/laccase hydrogel seems like an impractical approach to strengthening low cost cellulosic products. However, assuming the properties summarized in Table 2 for the PVAm-T10/laccase hydrogel layer and assuming the specific surface area of wood pulp fiber is 10 m^2/g, the mass of laccase required is 0.5% wt/wt based on dry pulp. The corresponding PVAm-T10 dose is only 0.3% wt/wt. These dosages are typical of conventional papermaking additives. However, if we consider only the TEMPO moiety dosage, the required dosage of TEMPO, tethered to PVAm, is only 0.0004% wt/wt based on pulp which is ten times less than Saito's recipe for cellulose oxidation with untethered TEMPO. [13] In summary, by concentrating the TEMPO and laccase in a thin adsorbed hydrogel layer, it is possible to achieve high local reagent concentrations with low dosages based on the mass of the cellulose.

CONCLUSIONS

We believe that PVAm-T/laccase complexes are the first reagentless cellulose grafting approach that functions at neutral pH and room temperature. The current work has focused on the redox mechanism and the main conclusions from this work are:

1. Adsorbed PVAm-T/laccase complexes formed a hydrogel layer typically 7 nm thick, 90 % water, containing laccase (200 mM) and TEMPO moieties (33 mM).
2. The density of cellulosic aldehyde groups on cellulose surfaces, generated by adsorbed PVAm-T/laccase, increases monotonically with the density of redox-active TEMPO moieties. Both the aldehyde and redox-active TEMPO densities were in the range 0.05 to 0.9 µmol/m^2.
3. The percolation threshold for efficient TEMPO-TEMPO electron transport within PVAm-T/laccase complexes corresponds to a TEMPO DS of 10% in PVAm-T. This

threshold corresponds to minimum TEMPO content required for high wet adhesion reported previously. [1]

4. Mixtures of PVAm-T and laccase display classic behaviors of polyelectrolyte complexes. When PVAm-T was in excess, the colloidal complexes were stable and positively charged. Similarly, if laccase was in excess, the colloidal complexes were stable and negatively charged.
5. Most of our experiments were performed using PVAm-T/laccase ratios near charge neutralization where the complexes were not colloidally stable. The aggregation of colloidal PVAm-T/laccase complexes resulted in a 50% decrease of oxidative capacity over the first two hours aging at pH 5.

ASSOCIATED CONTENT

***s Supporting Information**

The Supporting Information is available free of charge on the ACS Publications website at DOI:

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Notes

The authors declare no competing financial interests

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